

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026

Biology — Class IX | Paper: SSC-I

Syllabus: Ch 1 — The Science of Biology | Ch 2 — Biodiversity | Ch 3 — The Cell | Ch 4 — Cell Cycle | Ch 5 — Tissues, Organs & Organ Systems | Ch 6 — Molecular Biology | Ch 7 — Metabolism (Sec 7.1-7.6)

Total Marks: 65

Time: 2h 30m

Version: 3

General Instructions:

1. Write your Roll Number in the space provided on the answer sheet. Do NOT write your name.
2. Attempt ALL questions from Section A. In Sections B and C, attempt the required number choosing either the main question OR its alternative (OR).
3. All parts of a question must be answered in sequence and at one place.
4. Extra attempt of any question or any part thereof will NOT be considered.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question	A	B	C	D	Answer
1	The branch of biology concerned with the study of drugs such as aspirin, morphine, insulin, and penicillin is called:	☐ Pathology	☐ Physiology	☐ Immunology	☐ Pharmacology	
2	A mule, born from a male donkey and a female horse, is not considered a true biological species because it:	☐ Cannot survive alone	☐ Odd chromosome number	☐ No common ancestor	☐ Cannot be classified	
3	Which historical figure described 350 species of animals as early as the 700s, well before Linnaeus?	☐ Abu Usama Aljahiz	☐ Aristotle	☐ Andrea Caesalpino	☐ John Ray	
4	Lysosomes are sometimes described as "suicidal bags" of the cell because they contain enzymes that can:	☐ Produce ATP	☐ Store starch and lipids	☐ Synthesize new proteins	☐ Digest unwanted organelles (autophagy)	
5	During the G1 phase of interphase, the cell mainly:	☐ Replicates its DNA	☐ Forms the spindle apparatus	☐ Divides into two daughter cells	☐ Grows, prepares for DNA synthesis	
6	Which of the following is NOT one of the four phases of karyokinesis (nuclear division) in mitosis?	☐ Prophase	☐ Synapsis	☐ Metaphase	☐ Telophase	
7	Which organ system includes the pituitary, thyroid, parathyroid, pancreas, and adrenal glands, and communicates mainly through hormones?	☐ Endocrine system	☐ Nervous system	☐ Lymphatic system	☐ Integumentary system	
8	Which test is used to detect the presence of proteins in a solution, turning it pink or violet in colour?	☐ Iodine test	☐ Benedict's test	☐ Biuret test	☐ Emulsion test	
9	Cholesterol, an important component of the cell membrane, belongs to which class of lipids characterized by a three-ring structure and no fatty acids?	☐ Fatty acids	☐ Phospholipids	☐ Steroids	☐ Waxes	
10	Amylase, a starch-digesting enzyme, will NOT act on proteins or fats. This property of enzymes is called:	☐ Enzyme regulation	☐ Enzyme specificity	☐ Cofactor requirement	☐ Active site saturation	
11	Pepsin, which works inside the stomach cavity outside the cells that produced it, is an example of a(n):	☐ Intracellular enzyme	☐ Coenzyme	☐ Extracellular enzyme	☐ Ribozyme	
12	Without the presence of enzymes, a typical biochemical reaction in the body may take:	☐ Months or years	☐ Same time as with enzymes	☐ A few seconds	☐ A few milliseconds	

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt 11 questions. Choose either the main question OR its alternative.

Part	Question	Marks	OR	Alternative Question	Marks
2(i)	State any three Quranic references mentioned in Chapter 1 relating to the origin and development of life, along with the Sura name	3	OR	Name any three Muslim scientists mentioned in Chapter 1, and state the three major fields into which biology is divided.	3

	for each.				
2(ii)	State the four aims of classification mentioned in the textbook.	3	OR	Differentiate between autotrophic, heterotrophic, and saprotrophic modes of nutrition, giving one example organism for each.	3
2(iii)	Differentiate between prions and viroids as acellular infectious agents.	3	OR	Write the taxonomic hierarchy (from Kingdom to Species) for the classification of a Pea plant, as given in the textbook.	3
2(iv)	Describe the three layers of a plant cell wall and state the function of plasmodesmata.	3	OR	Describe the structure of the plant cell vacuole. What is the tonoplast, and how does the vacuole help maintain turgor pressure?	3
2(v)	Describe any three specialized cell types mentioned in the textbook (e.g. red blood cells, neurons, muscle cells) and explain how each cell's structure suits its function.	3	OR	Describe the structure and function of ribosomes in a cell. How do prokaryotic and eukaryotic ribosomes differ?	3
2(vi)	Down syndrome occurs when a person has 47 chromosomes instead of the usual 46, due to an extra copy of chromosome 21. Using your understanding of meiosis, explain what might go wrong during cell division to cause this.	3	OR	Define chromatid and centromere. Explain how sister chromatids are formed during the S phase of interphase.	3
2(vii)	Name the organs included in the lymphatic and immune systems, and state one homeostatic role of each.	3	OR	How do the muscular and integumentary systems contribute to maintaining body temperature (homeostasis)?	3
2(viii)	Give two examples of how the body works to restore homeostasis when disturbed, as mentioned in the textbook (during exercise and after drinking fruit juice).	3	OR	What organs make up the cardiovascular and skeletal systems, and what aspect of homeostasis does each regulate?	3
2(ix)	What organs make up the nervous and respiratory systems, and what aspect of homeostasis does each regulate?	3	OR	What is homeostasis? Name any two variables of the internal environment (other than temperature) that the body keeps stable, with one example each.	3
2(x)	Describe the structure of an amino acid, the building block of proteins, including the four groups attached to the central carbon atom.	3	OR	List the ten minor (trace) bioelements that make up about 1% of protoplasm.	3
2(xi)	What is a metabolic pathway? Explain the role of enzymes in forming this pathway, according to the textbook's definition.	3	OR	Define catabolism and anabolism. Give one example of each (other than photosynthesis or respiration), and state the relationship between energy release and energy use in these two processes.	3

SECTION C — Long Questions (4 × 5 = 20 Marks)

Part	Question	Marks	OR	Alternative Question	Marks
3(i)	Name any FIVE sub-fields of biology (other than Anatomy and Pharmacology) and give one example/application of each, as described in the textbook.	5	OR	Describe the relationship of biology with other sciences under the headings Biophysics, Biochemistry, Biostatistics, Computational Biology, and Biogeography, giving the focus of each as described in the textbook.	5
3(ii)	Describe the structure and functions of centrioles and the three types of cytoskeletal fibres (microtubules, microfilaments, intermediate filaments) found in a cell.	5	OR	Compare the structural advantages of plant and animal cells (any five points), and describe the concept of stem cells with reference to haematopoietic stem cells.	5
3(iii)	Using a comparison table, give any EIGHT differences between mitosis and meiosis.	5	OR	Describe how the spindle apparatus is formed during cell division (mention the role of centrioles and the special case of plant cells). Then explain the significance of mitosis in regeneration, asexual reproduction, and cloning/tissue culture — giving examples for each.	5
3(iv)	Describe the structure of lipids (fatty acids and glycerol) and differentiate between saturated and unsaturated fatty acids with examples. State any TWO functions of lipids in the body.	5	OR	Classify lipids into fats/oils, waxes and steroids with one example of each. Describe the tests used to detect the presence of lipids (spot test, water test, emulsion test).	5

Invigilator's Signature: _____

Candidate's Signature: _____

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— End of Question Paper —